

ARANSAS COUNTY AP, TX, USA

WMO: 722524

Lat: 28.084N

Lon: 97.046W

Elev: 22

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 12972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.8	38.9	21.2	15.9	47.0	26.0	20.0	48.5	25.8	50.9	23.9	51.0	11.8	0	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	93.1	78.6	91.9	78.9	91.0	78.9	81.5	88.6	81.0	88.1	80.4	87.4	11.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	153.0	84.2	79.2	151.9	84.2	78.9	149.9	84.2	45.1	89.0	44.5	87.8	43.9	87.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.8	18.2	DB	30.8	97.8	3.4	3.5	28.4	100.3	26.3	102.4	24.4	104.4	21.9	107.0	
			WB	28.0	82.6	3.7	1.0	25.4	83.3	23.2	83.9	21.2	84.4	18.5	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.7	57.1	60.8	66.3	72.7	78.9	83.8	84.7	85.4	82.2	75.4	66.0	58.8
	DBStd	11.96	8.92	8.67	7.54	5.71	3.90	2.52	2.14	2.09	4.00	6.93	8.46	8.85
	HDD50	86	37	17	5	0	0	0	0	0	0	0	5	22
	HDD65	861	270	163	78	13	0	0	0	0	0	12	91	234
	CDD50	8382	258	320	509	681	895	1013	1074	1098	966	789	485	294
	CDD65	3681	25	46	117	244	430	563	609	633	516	336	121	41
	CDH74	37663	58	82	307	1214	3774	6760	7691	8152	5894	3042	594	95
	CDH80	14601	8	7	30	187	949	2781	3415	3887	2431	843	61	2
(14) Wind	WSAvg	10.1	9.5	10.1	11.0	12.0	12.0	10.8	9.7	9.2	9.0	9.4	9.1	9.2
(15) Precipitation	PrecAvg	33.3	2.1	2.0	2.5	2.3	3.5	3.0	2.8	2.7	5.3	3.5	2.3	1.9
	PrecMax	48.2	6.7	6.3	7.2	8.7	11.5	9.7	17.9	8.6	15.6	11.7	9.4	6.2
	PrecMin	16.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.3	1.1	0.0	0.0	0.5
	PrecStd	9.5	1.7	1.7	1.9	2.0	3.1	2.3	4.2	1.9	3.4	2.9	2.4	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.4	79.9	83.1	88.4	90.1	93.5	93.3	95.4	95.3	90.5	83.9	79.3
		MCWB	65.0	67.3	67.2	67.6	75.7	76.7	79.0	78.7	77.1	77.5	74.8	70.7
	2%	DB	74.9	76.3	79.4	84.0	88.1	91.3	91.9	93.3	92.1	88.3	81.8	76.7
		MCWB	66.7	68.2	70.4	72.1	76.5	78.9	79.1	79.3	78.3	77.7	73.1	69.4
	5%	DB	72.4	73.7	77.4	81.8	86.4	90.4	91.0	92.2	90.7	86.4	80.0	74.3
		MCWB	65.7	67.2	69.0	72.3	76.3	78.8	79.1	79.4	78.4	76.5	71.4	68.5
	10%	DB	70.0	71.9	75.1	80.2	84.6	89.6	90.2	91.1	89.6	84.3	77.6	72.2
		MCWB	64.9	66.3	68.4	71.8	75.7	78.7	79.2	79.4	78.2	75.2	71.0	67.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	71.8	73.5	76.9	80.3	82.4	81.8	82.4	82.2	81.1	76.8	73.4
		MCDB	74.2	76.2	79.2	82.5	86.0	88.5	88.9	90.6	89.0	86.3	81.2	77.3
	2%	WB	69.3	70.4	72.3	75.7	78.9	81.2	81.0	81.5	81.0	79.8	75.2	71.9
		MCDB	72.2	73.9	76.9	80.9	84.4	87.1	88.4	89.7	87.9	84.7	79.7	75.0
	5%	WB	67.8	69.3	71.3	74.8	78.1	80.4	80.4	80.9	80.3	78.5	74.0	70.3
		MCDB	70.6	72.2	75.1	79.6	83.7	86.6	87.7	88.8	87.0	83.7	78.1	73.2
	10%	WB	66.2	68.0	70.4	73.8	77.4	79.8	80.0	80.3	79.6	77.6	72.7	68.6
		MCDB	68.9	70.5	73.7	78.1	83.1	86.3	87.4	88.0	86.3	83.1	76.6	71.5
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	14.6	13.8	13.0	11.6	11.5	11.3	12.6	13.6	14.7	16.0	16.2
		MCDBR	17.4	15.0	13.3	13.5	12.6	12.4	12.1	13.6	14.4	13.2	14.9	16.4
		MCWBR	11.7	9.3	7.0	6.7	5.3	4.4	4.1	4.4	5.3	6.3	8.5	11.5
	5% WB	MCDBR	14.6	12.6	11.2	10.5	10.1	10.6	11.0	12.1	12.3	11.7	13.4	14.6
		MCWBR	11.3	9.3	6.7	5.7	4.7	4.6	4.4	4.5	5.3	6.8	9.3	11.8
(40) Clear-Sky Solar Irradiance	taub		0.359	0.367	0.386	0.432	0.476	0.451	0.469	0.450	0.441	0.393	0.379	0.348
	taud		2.484	2.466	2.409	2.287	2.196	2.330	2.295	2.349	2.338	2.466	2.453	2.534
	Ebn at Noon		280	287	287	275	262	267	262	267	266	275	270	279
	Edh at Noon		30	33	37	43	47	41	42	40	39	32	30	27
(44) All-Sky Solar Radiation	RadAvg		942	1111	1439	1712	1930	2124	2086	1995	1653	1438	1078	850
	RadStd		104	183	146	161	134	114	180	109	143	148	115	111

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153

Nomenclature: See separate page